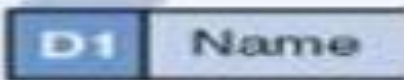


DATA FLOW DIAGRAMS DFD's



GA NE & SARSON SYMBOLS

For DFDs we are going to use Gane and Sarson symbols in our projects

Item	Gane & Sarson Symbol
External Entity	
Process	
Data store	
Data flow	

DFD NOTATIONS

Process Notations - transforms incoming data flow into outgoing data flow.

Data store Notations - repositories of data in the system, sometimes files.

Dataflow Notations – labelled pipelines through which named packets of information flow.

External Entity Notations - objects outside the system, with which the system interacts with. They are sources and destinations of the system's inputs and outputs.

RULES OF DATA FLOWS

One of the rules for developing DFD is that all data flow must begin with and end at a processing step.

Data flows from an Entity into a process

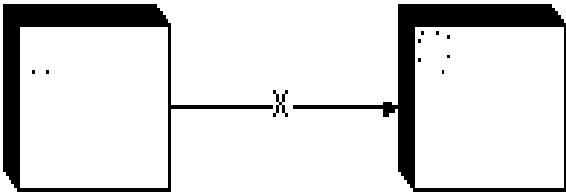
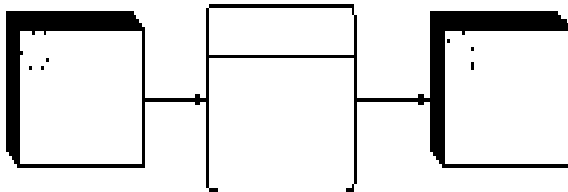
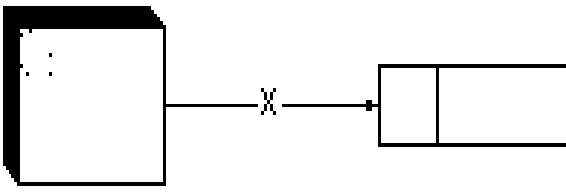
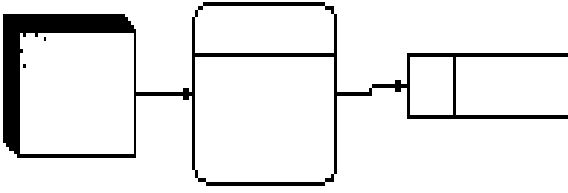
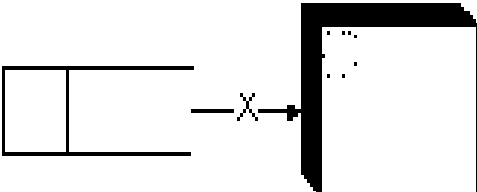
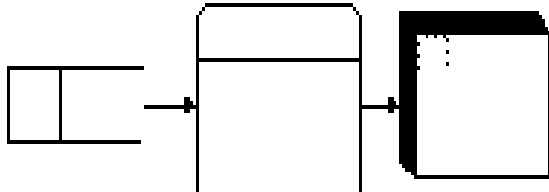
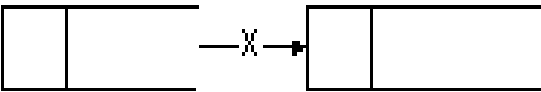
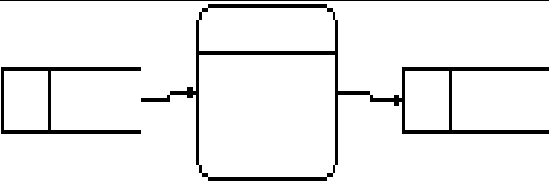
Data flows from a process to another process

Data flows from a process to a data store

COMMON MISTAKES IN DFDs

- **Black holes** - A processing step with input flows but no output flows.
- **Miracles** - A processing step with output flows but no input flows.
- **Grey holes** - A processing step with outputs that are greater than the sum of its inputs

Common mistakes (cont'd)

Wrong	Right	Description
		A source or a sink cannot provide data to another source or sink without some processing occurring.
		Data cannot move directly from a source to a data store without being processed.
		Data cannot move directly from a data store to a sink without being processed.
		Data cannot move directly from one data store to another without being processed.



DATA STORE

It is used in a DFD to represent a situation when the system must retain data because one or more processes need to use the stored data later on in the life of the system.



DATA STORE CONT'D



Data can be written into the data store, which is depicted by an outgoing arrow from a process



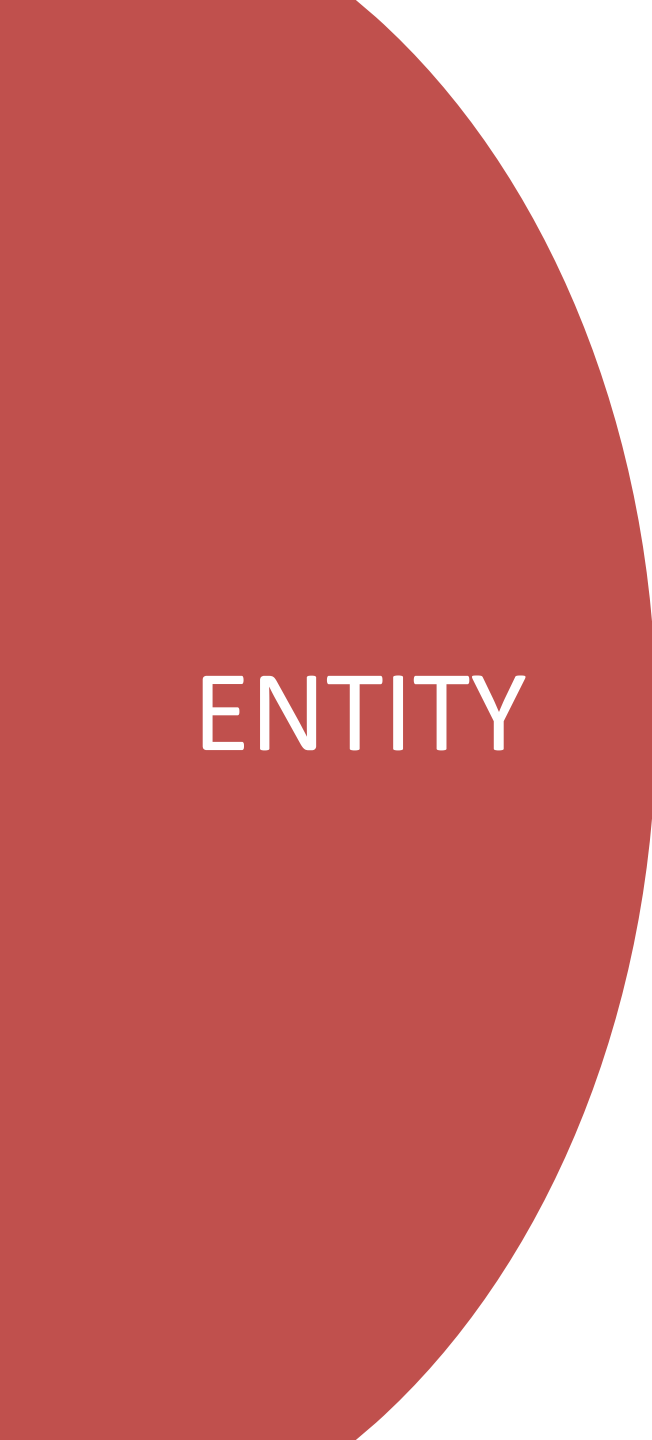
Data can be read from a data store, which is depicted by an incoming arrow into a process.



A data store must be connected to a process with a dataflow.

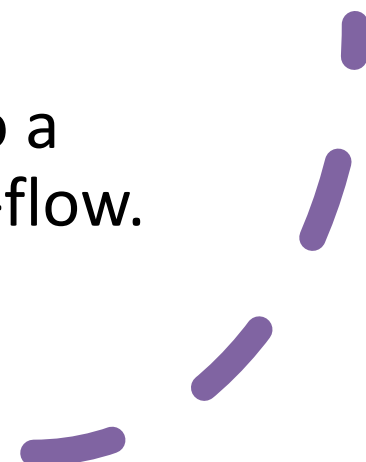


Each data store must have at least one input dataflow and at least one output dataflow. The dataflow can either be a control or confirmation message.



ENTITY

They represent how the information system interacts with the outside world.

- A rectangle represents an external entity
 - They either supply data or receive data (sources or sinks)
 - They do not process data
 - Also terminators because they are data origins or final destinations.
 - It must be connected to a process through a data-flow.
- 

DFD DEV GUIDELINES

CONTEXT DIAGRAM - LEVEL 0

- must fit on one page
- process name should be the name of the information system eg Registration System.
- gets the process number 0 (level zero)
- No data stores

UNIQUE NAME FOR LEVELS - use unique names within each set of symbols no duplicate names

NO CROSS LINE IN DFD - achieved by restricting number of processes in a DFD

DFD DEV GUIDELINES CONT'D

NUMBERING CONVENTION

Use a unique reference number for each process symbol.

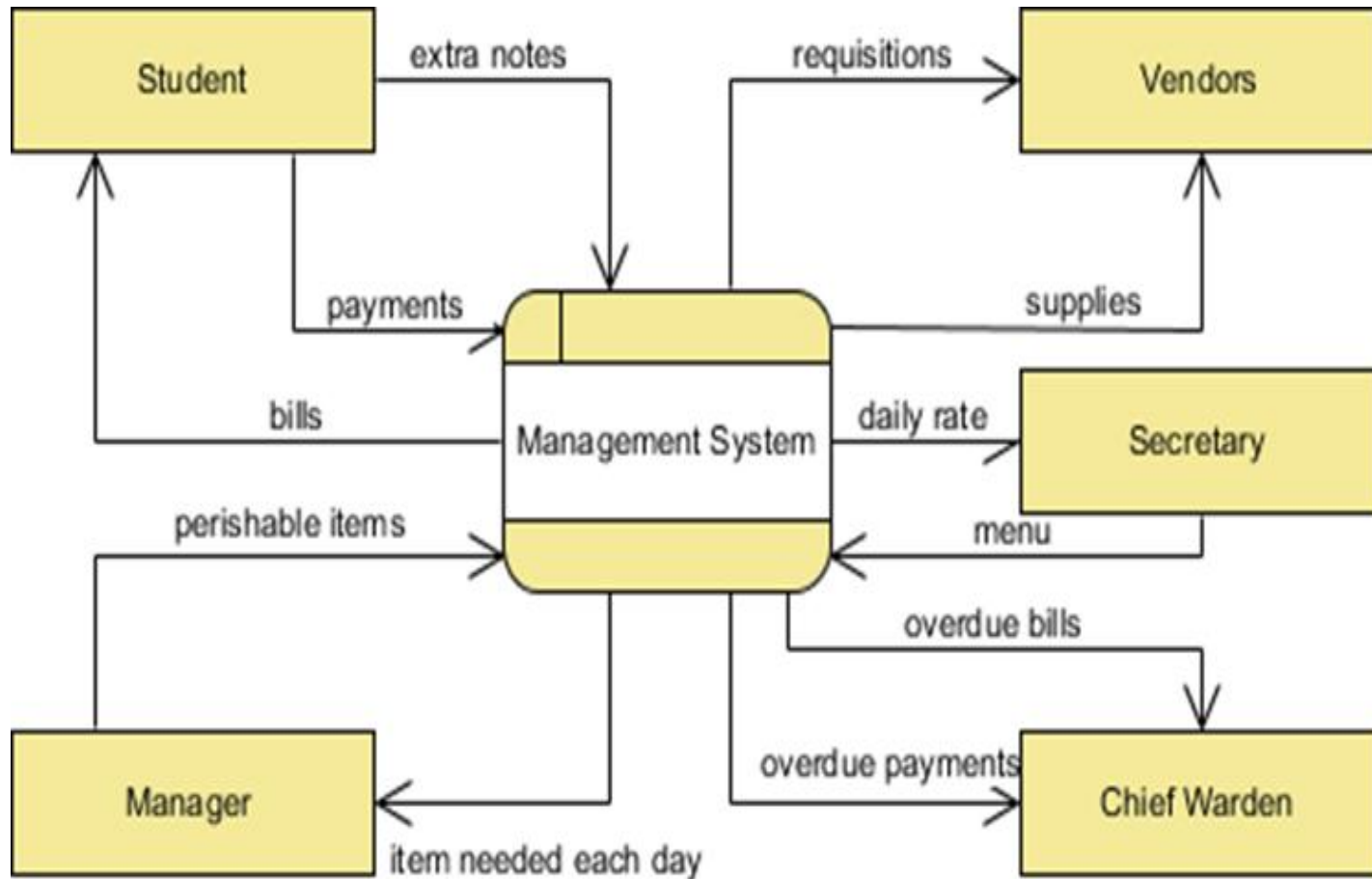
Other lower level process numbers are in the hierarchy of:

- (1, 2, 3,...); - major processes usually derived from objectives
- (1.1, 1.2, 1.3, ..., 2.1, 2.2, 2.3,...); - sub-processes
- (1.1.1, 1.1.2, 1.1.3,...) – sub sub-processes

Context diagrams

- A context diagram is a graphical representation of a system which must only use one process to represent the entire system and deliberately does not go into defining all the processes so as to prevent people getting bogged down in complex details at an early stage. There are only three symbols used in a context diagram:
 - A circle/rounded square to represent the system in terms of a single process. There will never be more than a single process in any context diagram,
 - arrows to represent data flow,
 - and a rectangle to represent any external entities affecting the system. There can be numerous external entities.
- Many students find Context Diagrams confusing because they are so much like a DFD. This is easy to explain because a context diagram is a level 0 DFD. This is the most basic type of DFD where all the processes and storage is represented by a single process. Context diagrams do not have storage. DFDs will always have storage.

CONTEXT LEVEL DFD EXAMPLE



Data flow diagrams

- A Data flow diagram (DFD) is a much more complex representation of a context diagram. DFD show a further level of detail not shown in the context diagram. DFD's identify the source of data, its flow between processes and its destination. It along also shows data generated by the system.
- Data flow diagrams represent an information system as a number of processes that together form the single system. As mentioned earlier, a data flow diagram is a more detailed form of a context diagram (see above). It will show processes, inputs, outputs and storage. The Data Flow Diagram (DFD) shows the data flow between the processes within a system. A DFD can become as detailed as the user requires. The DFD will become increasingly detailed as the levels go higher. However each level will tend to focus on extending one or more of the processes rather than the entire system.
- The first level DFD shows the main processes within the system. Each of these processes can be broken into further processes.

LEVEL 1 DFD EXAMPLE

